

Autonomous Physics-Based Modeling of Retention Loss in 3-D NAND Flash Memories

Myung Jin and , *Member, IEEE*

Abstract—A novel model is established with physical derivation in retention operation of 3-D NAND flash memory. To efficiently incorporate the retention loss properties of the immediate and continuous occurrence, an autonomous model has been suggested. By utilizing the model, retention loss for both short-term and long-term due to detrapping effect of trapped electron can be simulated in very short timeframe. Moreover, autonomous model makes it possible for model to predict the time transient of ΔV_{th} by only using an initial condition.

Index Terms—Bandgap-Engineered 3D NAND Flash, Soft Program, Pass Voltage, ΔV_T Distribution

I. INTRODUCTION

THERE is a rapid increase in the demand for data storage driven by modern computing technologies such as artificial intelligence (AI) and high-performance computing (HPC), 3-D NAND flash memories' reliability issue for retention loss must be concerned. As data storage capacity per cell increases, the reliability of each bit's ΔV_{th} distribution must be maintained over a long period, ranging from a few days to several years.

It has been reported that detrapping effect occur under cryogenic condition. This aligns with the previous study of detrapping modeled independent of temperature.

In this study, an autonomous modeling of the retention loss by tunneling emission of 3-D NAND flash memory has been constructed. Only initial condition matters for this simulation with highly fast simulation time.

II. MODELING

The modeling of autonomous simulation for retention loss is held in this section. First, a cylindrical poisson solution for O/N/O layer will be done with quantized multiple layer in CTN layer. After the detailed poisson solution for multiple quantized layers is solved, ΔV_{th} will be solved with optimized calculation. Last, based on a previous study of emission rate by detrapping effect, an autonomous equation is suggested.

A. Detailed Poisson Solution

An electrical potential distribution of a cylindrical Poisson equation for split layers in ONO structure can be solved as

Myung Jin is with the Seoul National University(e-mail: gsjb1028@snu.ac.kr).

S. B. Author, Jr., was with Rice University, Houston, TX 77005 USA. He is now with the Department of Physics, Colorado State University, Fort Collins, CO 80523 USA (e-mail: author@lamar.colostate.edu).

following equation.

$$\begin{cases} V_{Si}(r_{Si}) = V_{ch} \\ V_{TOX}(r) = V_{ch} + \frac{K_{TOX}}{\epsilon_{TOX}} \ln \frac{r}{r_{Si}} \\ V_{CTN,i}(r) = V_{CTN,i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{CTN}} \ln \frac{r}{r_{i-1}} \\ V_{BOX}(r) = V_{CTN,N}(r_N) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_N} \\ V_{BOX}(r_{BOX}) = V_G \end{cases} \quad (1)$$

Solving $\mathbf{x}$ vector can calculate the total $N + 2$ potential parameters.

$$\mathbf{x} = [K_{TOX} \quad K_1 \quad \cdots \quad K_N \quad K_{BOX}]^T \quad (2)$$

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad (3)$$

$\mathbf{A}$ and $\mathbf{b}$ can be defined with total $N + 1$ transverse electric field equality boundary conditions ($\epsilon_1 \mathcal{E}_1 = \epsilon_2 \mathcal{E}_2$) and one extra boundary condition with reference voltage.

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{TOX}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{BOX}} \end{bmatrix} \quad (4)$$

$$\mathbf{b} = [\frac{\Delta Q_0}{2\pi} \quad \frac{\Delta Q_1}{2\pi} \quad \cdots \quad \frac{\Delta Q_N}{2\pi} \quad \Delta V]^T \quad (5)$$

$$\Delta V = (V_G - V_{ch}) - \sum_{i=1}^N \frac{qn_{t,i}}{4\epsilon_{CTN}} (r_i^2 - r_{i-1}^2) \quad (6)$$

C_{TOX} , C_i , and C_{BOX} in (4) indicates capacitance of each cylindrical layers as shown in Fig. 1(b)-(c). ΔQ_i in (5) stands for a compenet of radius r_i cylindrical charge which can be superpositioned as $Q = \sum \Delta Q_i$ as demonstrated on Fig. 1(d). (5) shows that $\mathbf{b}(\Delta Q_i, V_G - V_{ch})$ is varied by trapped charge distribution and given voltage conditions.

B. $Q - \Delta V_{th}$ Relation

For ΔV_{th} calculation, gate voltage difference for equal channel field condition should be solved.

$$F_{TOX}(r_{Si}) = \frac{K_{TOX}}{\epsilon_{TOX}} \frac{1}{r_{Si}} \quad (7)$$

to achieve equal channel field, same K_{TOX} value under given conditions must be established. By Cramer's rule, K_{TOX} parameter can be solved by determinant calculation.

$$K_{\text{TOX}} = \frac{\det [A_{\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})}]}{\det A} \quad (8)$$

where notation $A_{\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})}$ stands for 1st column of matrix A replaced with vector $\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})$.

Therefore, establishing following equation and solving for ΔV_{th} is needed to calculate the value.

$$\det [A_{\mathbf{b}(0, \Delta V_{\text{th}})}] = \det [A_{\mathbf{b}(\Delta Q_i, 0)}] \quad (9)$$

by property of linear conservation in determinant,

$$\det \begin{bmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \Delta V_{\text{th}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\text{BOX}}} \end{bmatrix} = -\Delta V_{\text{th}} \quad (10)$$

$$\therefore \Delta V_{\text{th}} = -\det [A_{\mathbf{b}(\Delta Q_i, 0)}] \quad (11)$$

The threshold voltage difference can be calculated with single determinant computation with initial trapped charge distribution is given.

C. Autonomous Detrapping Modeling

With given initial condition, an autonomous equation for each cell divided by radial and trap depth as Fig. 2(a) is established.

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}}) n_t(r, E_t, t) \quad (12)$$

Then, detrapping relation can be expressed as following general ODE solution.

$$n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0} \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \quad (13)$$

for hole concentration (h_t) vice versa.

The emission rate for trapped electron and hole can be computed as following equations based on a previous study [1].

$$e = \nu \times |TC| \quad (14)$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m^* \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}} \quad (15)$$

$$|TC| = \prod \exp \frac{4\sqrt{2m^*} \Phi_H^{1.5} - \Phi_L^{1.5}}{3\hbar q F} \quad (16)$$

for both channel and wordline directions as demonstrated in Fig. 2(a).

REFERENCES

- [1] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, 2nd ed., vol. 3, J. Peters, Ed. New York, NY, USA: McGraw-Hill, 1964, pp. 15–64.
- [2] W.-K. Chen, *Linear Networks and Systems*. Belmont, CA, USA: Wadsworth, 1993, pp. 123–135.
- [3] J. U. Duncombe, "Infrared navigation—Part I: An assessment of feasibility," *IEEE Trans. Electron Devices*, vol. ED-11, no. 1, pp. 34–39, Jan. 1959, 10.1109/TED.2016.2628402.
- [4] E. P. Wigner, "Theory of traveling-wave optical laser," *Phys. Rev.*, vol. 134, pp. A635–A646, Dec. 1965.
- [5] E. H. Miller, "A note on reflector arrays," *IEEE Trans. Antennas Propagat.*, to be published.
- [6] E. E. Reber, R. L. Michell, and C. J. Carter, "Oxygen absorption in the earth's atmosphere," Aerospace Corp., Los Angeles, CA, USA, Tech. Rep. TR-0200 (4230-46)-3, Nov. 1988.
- [7] J. H. Davis and J. R. Cogdell, "Calibration program for the 16-foot antenna," Elect. Eng. Res. Lab., Univ. Texas, Austin, TX, USA, Tech. Memo. NGL-006-69-3, Nov. 15, 1987.
- [8] *Transmission Systems for Communications*, 3rd ed., Western Electric Co., Winston-Salem, NC, USA, 1985, pp. 44–60.
- [9] *Motorola Semiconductor Data Manual*, Motorola Semiconductor Products Inc., Phoenix, AZ, USA, 1989.
- [10] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, vol. 3, Polymers of Hexadromicon, J. Peters, Ed., 2nd ed. New York, NY, USA: McGraw-Hill, 1964, pp. 15–64. [Online]. Available: <http://www.bookref.com>.
- [11] *The Founders' Constitution*, Philip B. Kurland and Ralph Lerner, eds., Chicago, IL, USA: Univ. Chicago Press, 1987. [Online]. Available: <http://press-pubs.uchicago.edu/founders/>
- [12] The Terahertz Wave eBook. ZOmega Terahertz Corp., 2014. [Online]. Available: http://dl.z-thz.com/eBook/zomega_ebook_pdf_1206_sr.pdf. Accessed on: May 19, 2014.
- [13] Philip B. Kurland and Ralph Lerner, eds., *The Founders' Constitution*. Chicago, IL, USA: Univ. of Chicago Press, 1987, Accessed on: Feb. 28, 2010, [Online] Available: <http://press-pubs.uchicago.edu/founders/>
- [14] J. S. Turner, "New directions in communications," *IEEE J. Sel. Areas Commun.*, vol. 13, no. 1, pp. 11–23, Jan. 1995.
- [15] W. P. Risk, G. S. Kino, and H. J. Shaw, "Fiber-optic frequency shifter using a surface acoustic wave incident at an oblique angle," *Opt. Lett.*, vol. 11, no. 2, pp. 115–117, Feb. 1986.
- [16] P. Kopyt *et al.*, "Electric properties of graphene-based conductive layers from DC up to terahertz range," *IEEE THz Sci. Technol.*, to be published. DOI: 10.1109/THZ.2016.2544142.
- [17] PROCESS Corporation, Boston, MA, USA. Intranets: Internet technologies deployed behind the firewall for corporate productivity. Presented at INET96 Annual Meeting. [Online]. Available: <http://home.process.com/Intranets/wp2.htm>
- [18] R. J. Hijmans and J. van Eten, "Raster: Geographic analysis and modeling with raster data," R Package Version 2.0-12, Jan. 12, 2012. [Online]. Available: <http://CRAN.R-project.org/package=raster>
- [19] Teralyzer. Lytera UG, Kirchhain, Germany [Online]. Available: <http://www.lytera.de/Terahertz-THz-Spectroscopy.php?id=home>, Accessed on: Jun. 5, 2014
- [20] U.S. House. 102nd Congress, 1st Session. (1991, Jan. 11). *H. Con. Res. 1, Sense of the Congress on Approval of Military Action*. [Online]. Available: LEXIS Library: GENFED File: BILLS
- [21] Musical toothbrush with mirror, by L.M.R. Brooks. (1992, May 19). Patent D 326 189 [Online]. Available: NEXIS Library: LEXPAT File: DES
- [22] D. B. Payne and J. R. Stern, "Wavelength-switched passively coupled single-mode optical network," in *Proc. IOOC-ECOC*, Boston, MA, USA, 1985, pp. 585–590.
- [23] D. Ebehard and E. Voges, "Digital single sideband detection for interferometric sensors," presented at the 2nd Int. Conf. Optical Fiber Sensors, Stuttgart, Germany, Jan. 2–5, 1984.
- [24] G. Brandli and M. Dick, "Alternating current fed power supply," U.S. Patent 4 084 217, Nov. 4, 1978.
- [25] J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, USA, 1993.
- [26] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.
- [27] A. Harrison, private communication, May 1995.
- [28] B. Smith, "An approach to graphs of linear forms," unpublished.

- [29] A. Brahms, "Representation error for real numbers in binary computer arithmetic," IEEE Computer Group Repository, Paper R-67-85.
- [30] IEEE Criteria for Class IE Electric Systems, IEEE Standard 308, 1969.
- [31] Letter Symbols for Quantities, ANSI Standard Y10.5-1968.
- [32] R. Fardel, M. Nagel, F. Nuesch, T. Lippert, and A. Wokaun, "Fabrication of organic light emitting diode pixels by laser-assisted forward transfer," *Appl. Phys. Lett.*, vol. 91, no. 6, Aug. 2007, Art. no. 061103.
- [33] J. Zhang and N. Tansu, "Optical gain and laser characteristics of InGaN quantum wells on ternary InGa_N substrates," *IEEE Photon. J.*, vol. 5, no. 2, Apr. 2013, Art. no. 2600111.
- [34] S. Azodolmolky *et al.*, Experimental demonstration of an impairment aware network planning and operation tool for transparent/translucent optical networks," *J. Lightw. Technol.*, vol. 29, no. 4, pp. 439–448, Sep. 2011.